

IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Nom: BELDOR

Prénom: Warkens Holmando

Niveau: L3

Date: 8/11/2026

Cours: Réseau1

Professeurs: M. SAINT AMOUR Ismaël

Objectif du TD

Ce travail pratique avait pour objectifs principaux de :

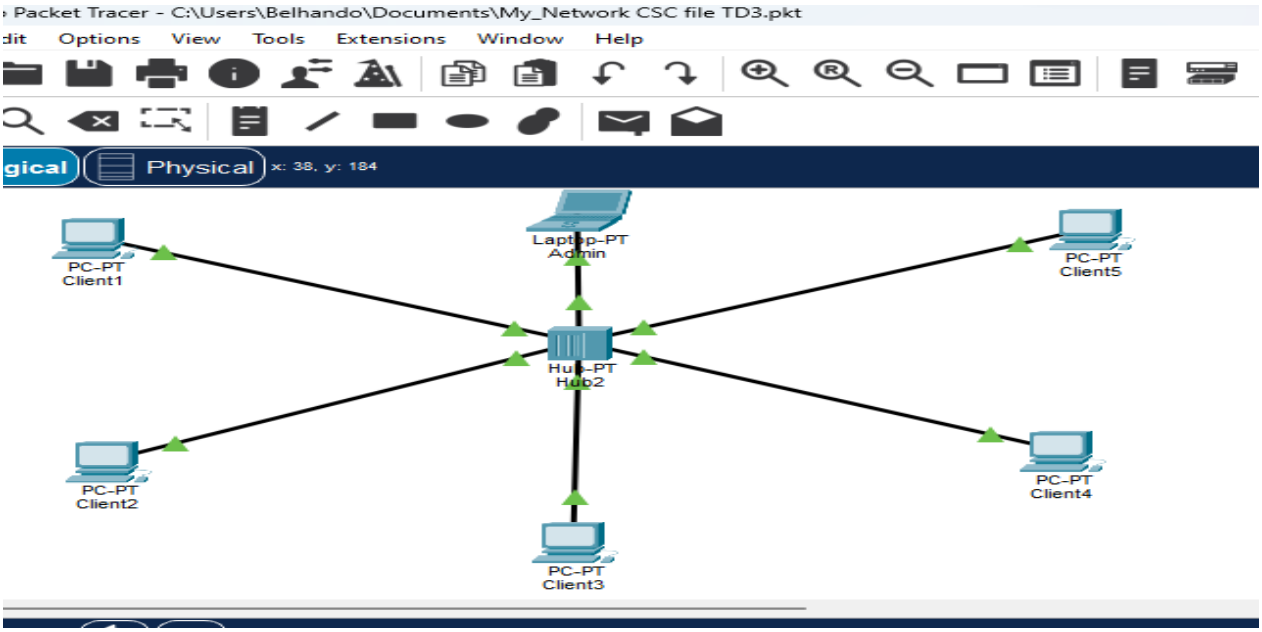
Savoir attribuer des adresses IP valides (IPv4 et IPv6) aux différentes machines d'un réseau.

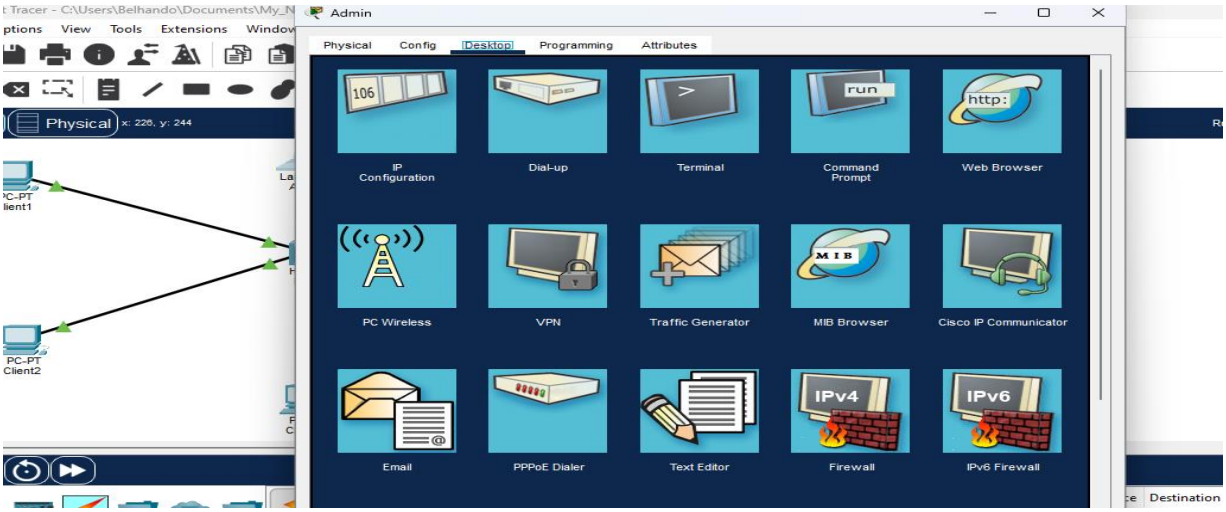
Comprendre le fonctionnement fondamental de l'adressage IPv4 et IPv6.

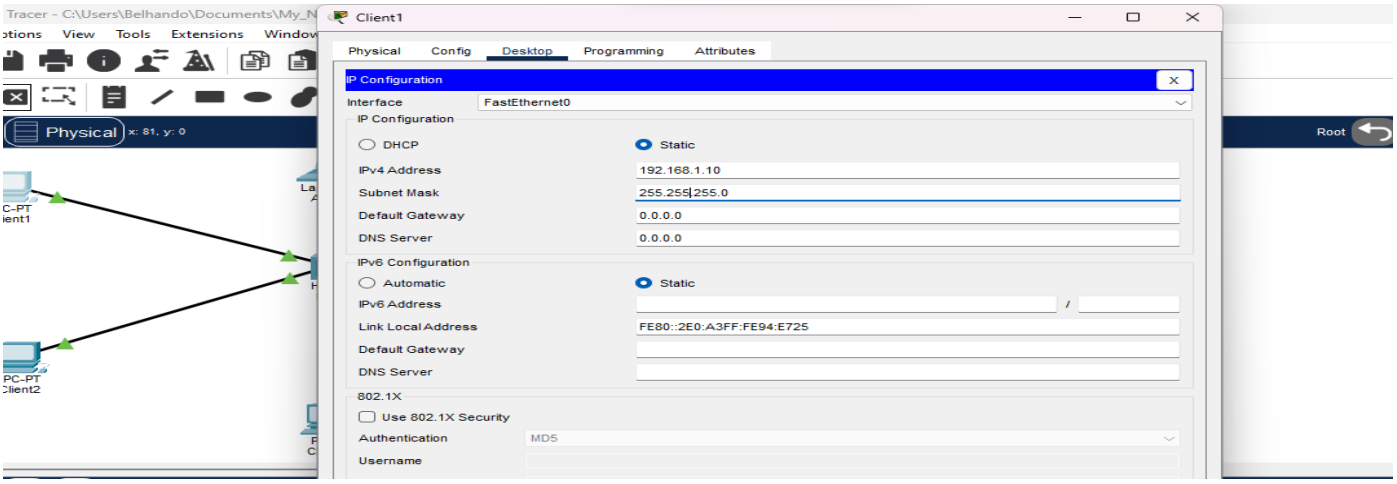
Configurer ces adresses IP sur les hôtes au sein du logiciel de simulation Cisco Packet Tracer.

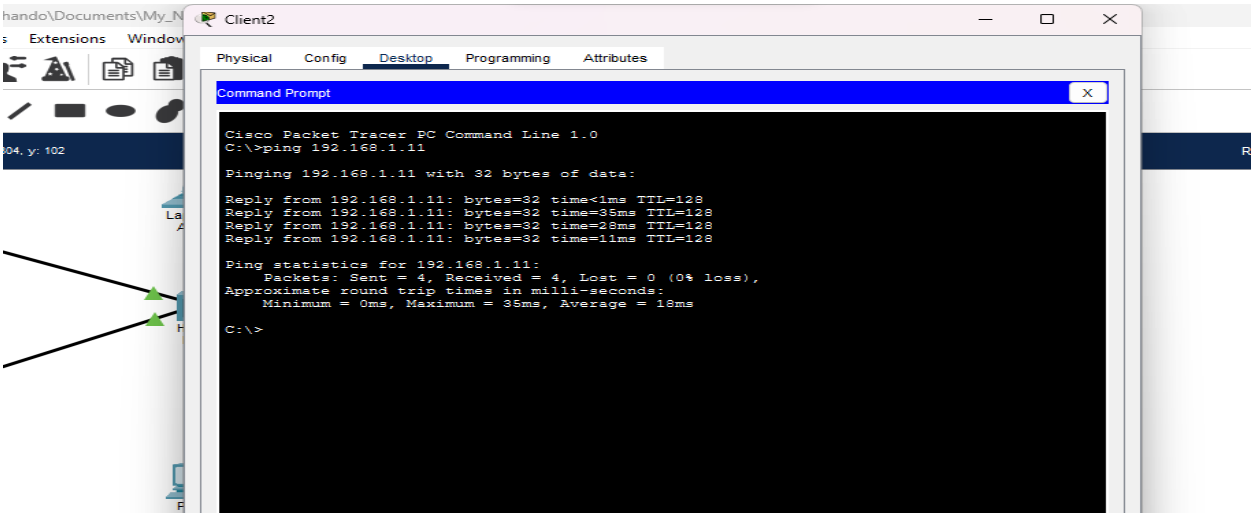
Vérifier et valider la connectivité réseau à l'aide des commandes ping (IPv4) et ping (IPv6).

1- Reproduction de topologie , configuration des addresses IPV4 et vérification de connectivité.

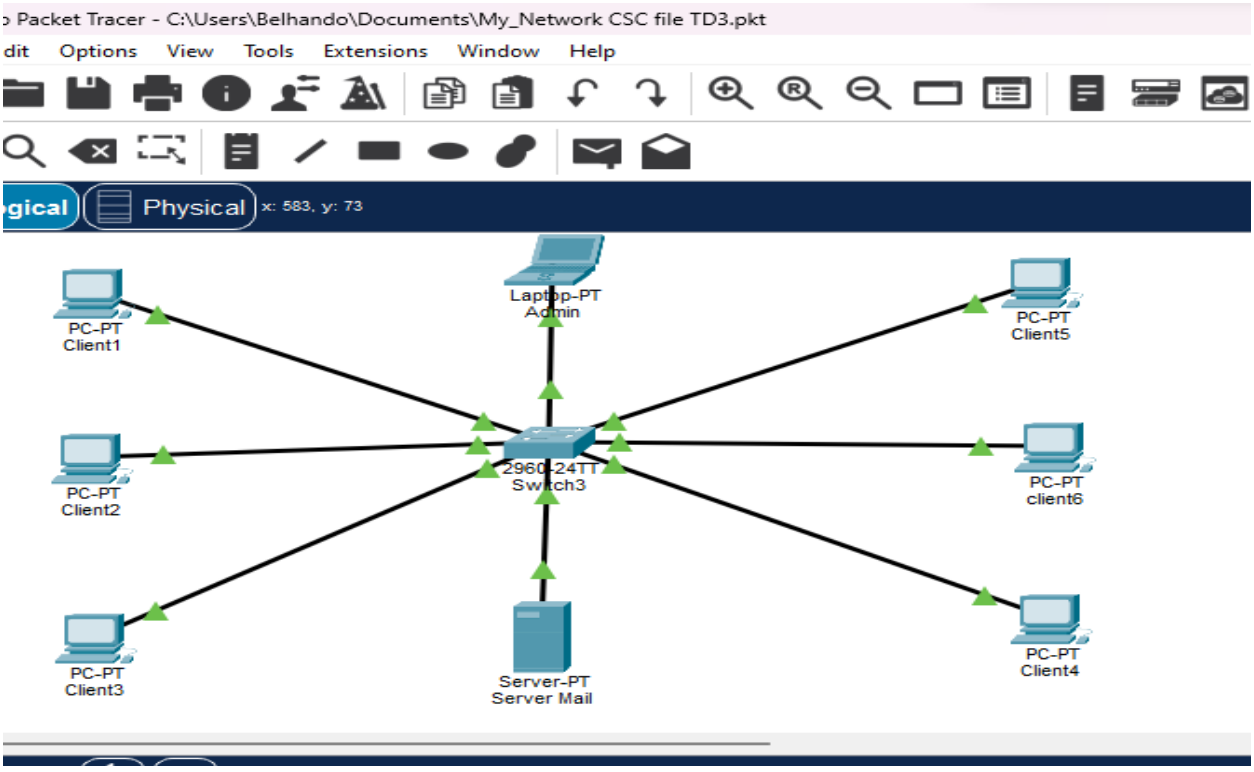


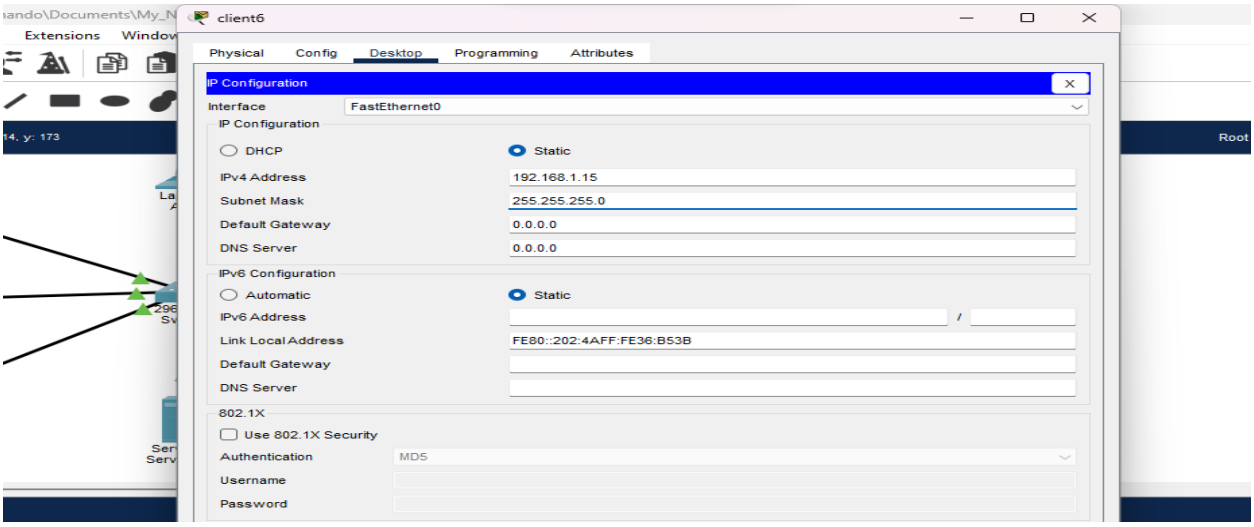

Admin
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Dial-up
Terminal
Command Prompt
Web Browser
PC Wireless
VPN
Traffic Generator
MIB Browser
Cisco IP Communicator
Email
PPPoE Dialer
Text Editor
Firewall
IPv6 Firewall
Physical
PC-PT Client2
Destination

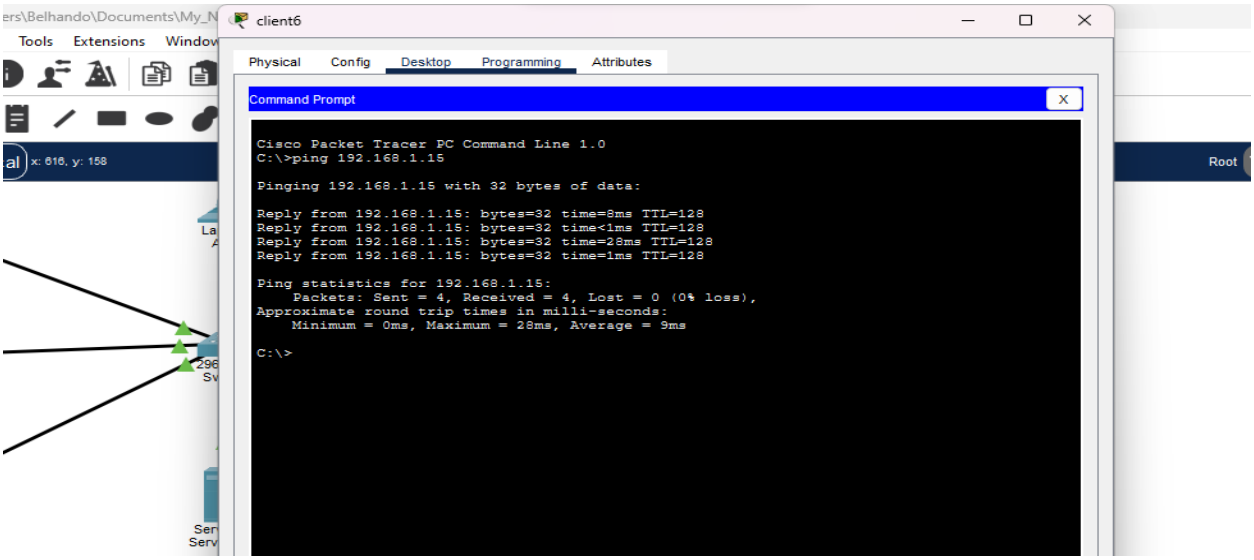

Client1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:A3FF:FE94:E725
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Physical
Root
PC-PT Client2

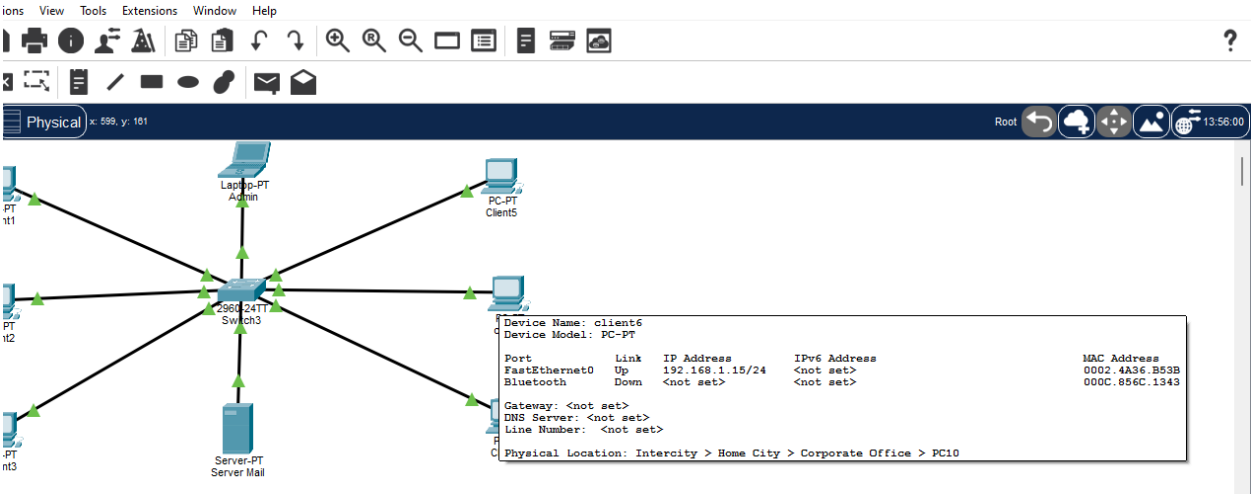


2- Reproduction de topologie, configuration des addresses IPV4 t vérification de connectivité.

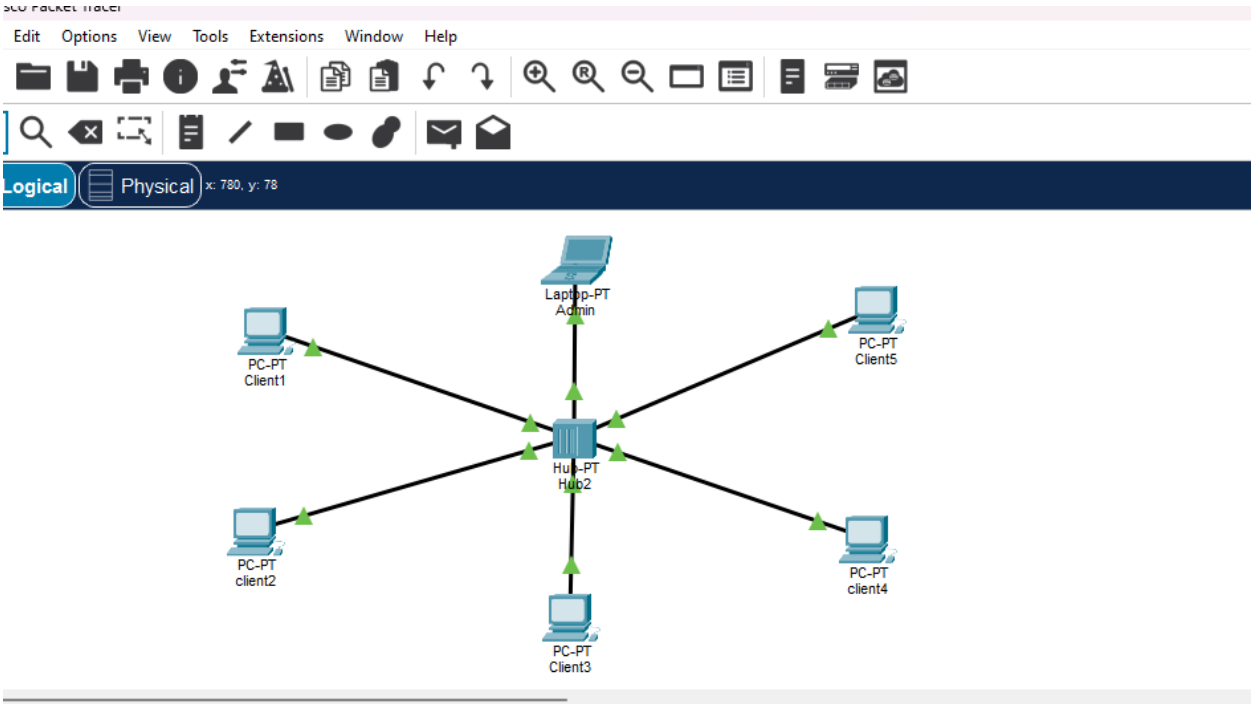


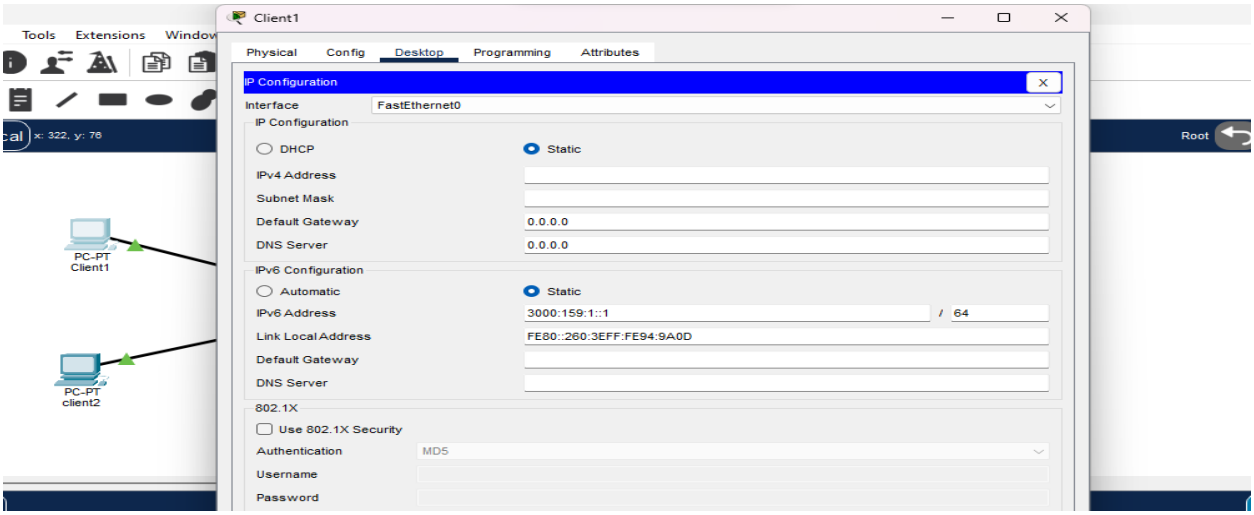

client6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.15
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:4AFF:FE36:B53B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

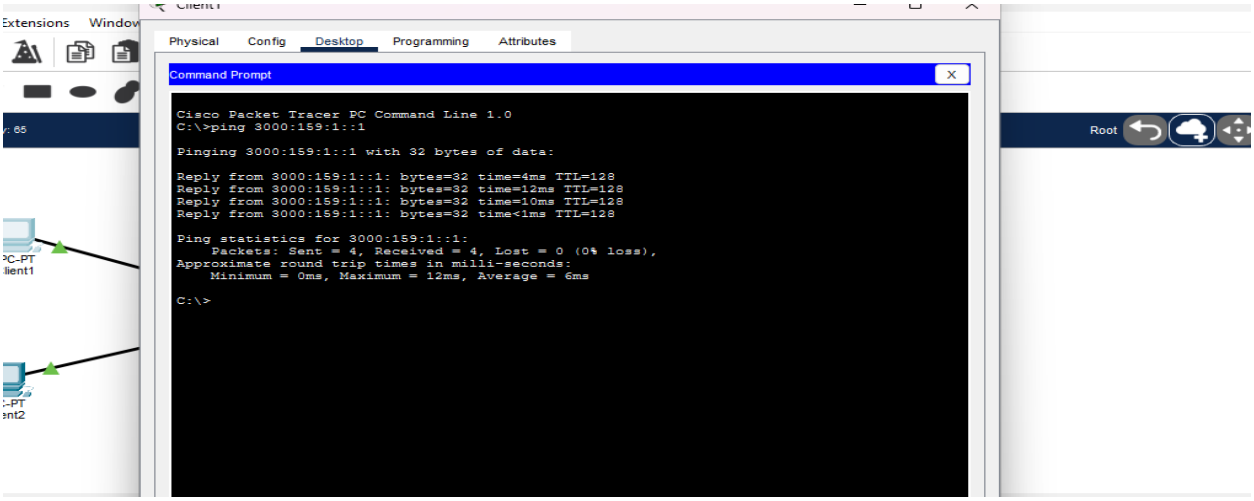

client6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time=8ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=28ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 28ms, Average = 9ms
C:\>

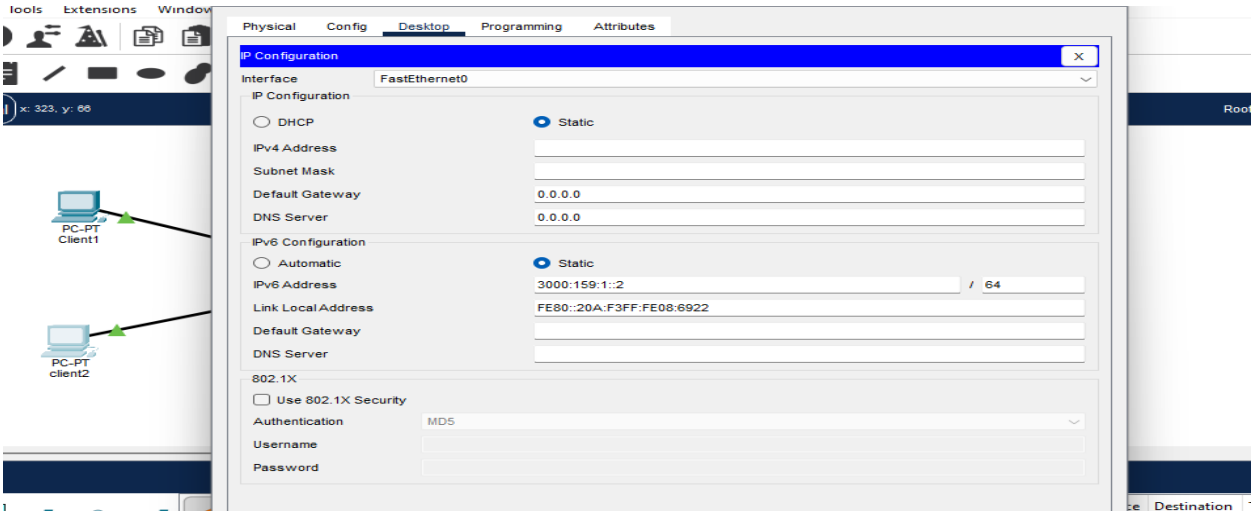


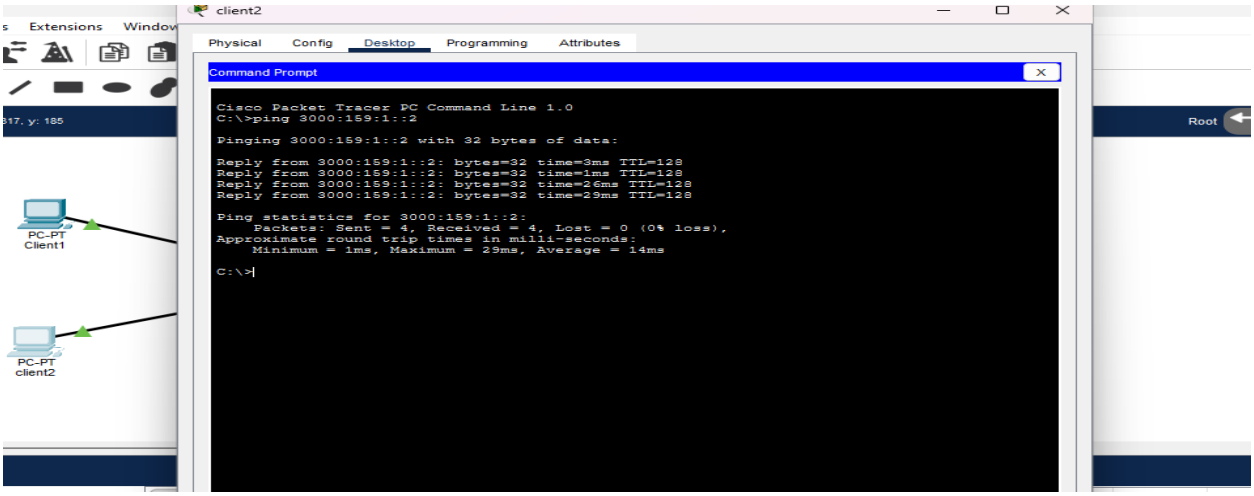
1- Reproduction de topologie, configuration des addresses IPV6 t vérification de connectivité.

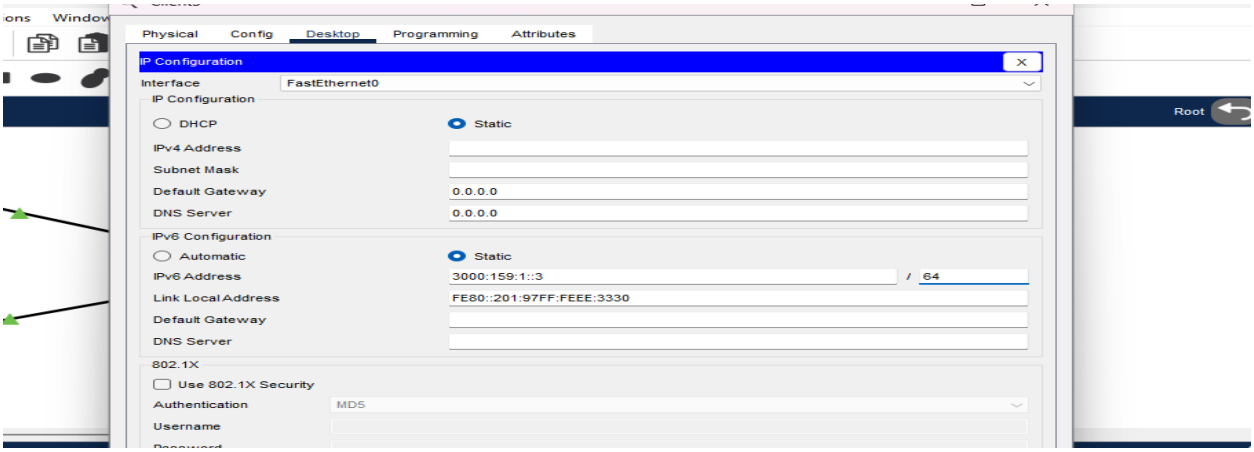


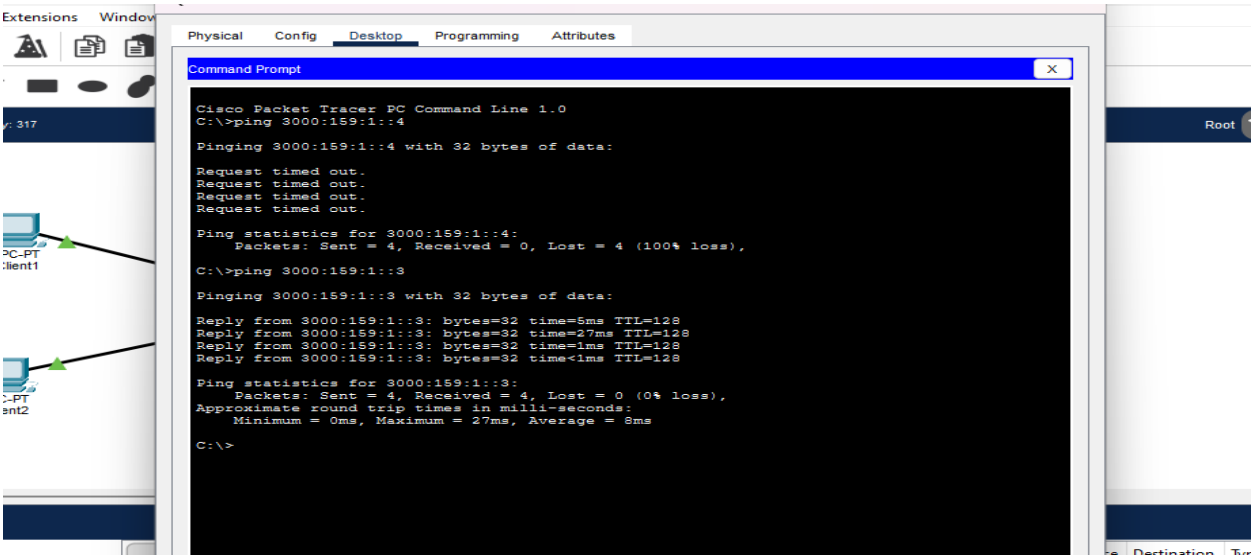

Client1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::1
/ 64
Link Local Address
FE80::260:3EFF:FE94:9A0D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
PC-PT Client1
PC-PT client2
Root

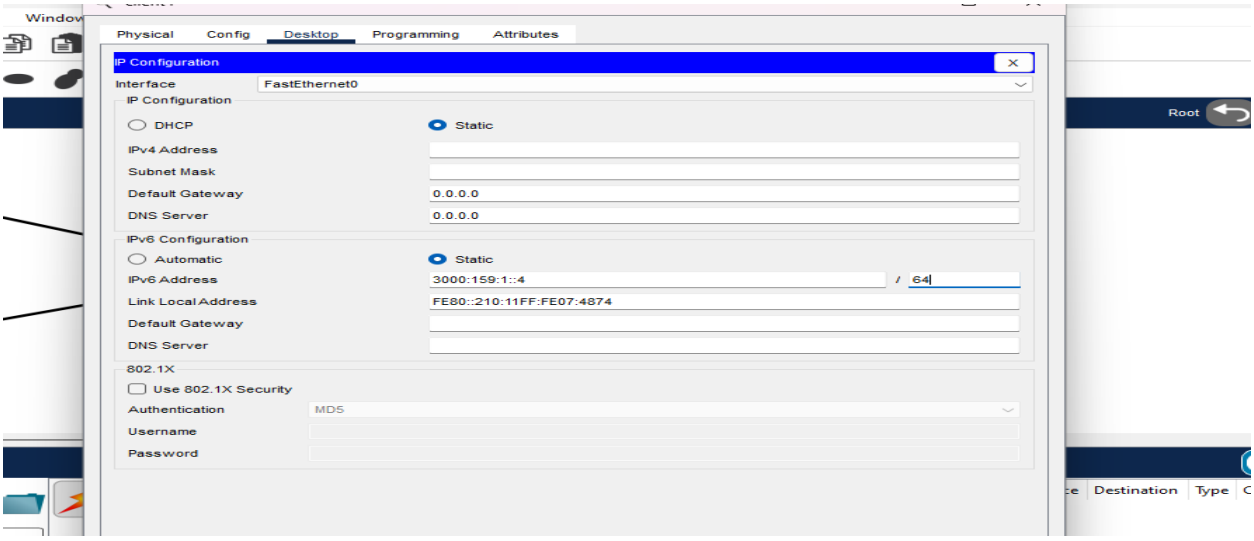

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::1
Pinging 3000:159:1::1 with 32 bytes of data:
Reply from 3000:159:1::1: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=12ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=10ms TTL=128
Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 6ms
C:\>
Root

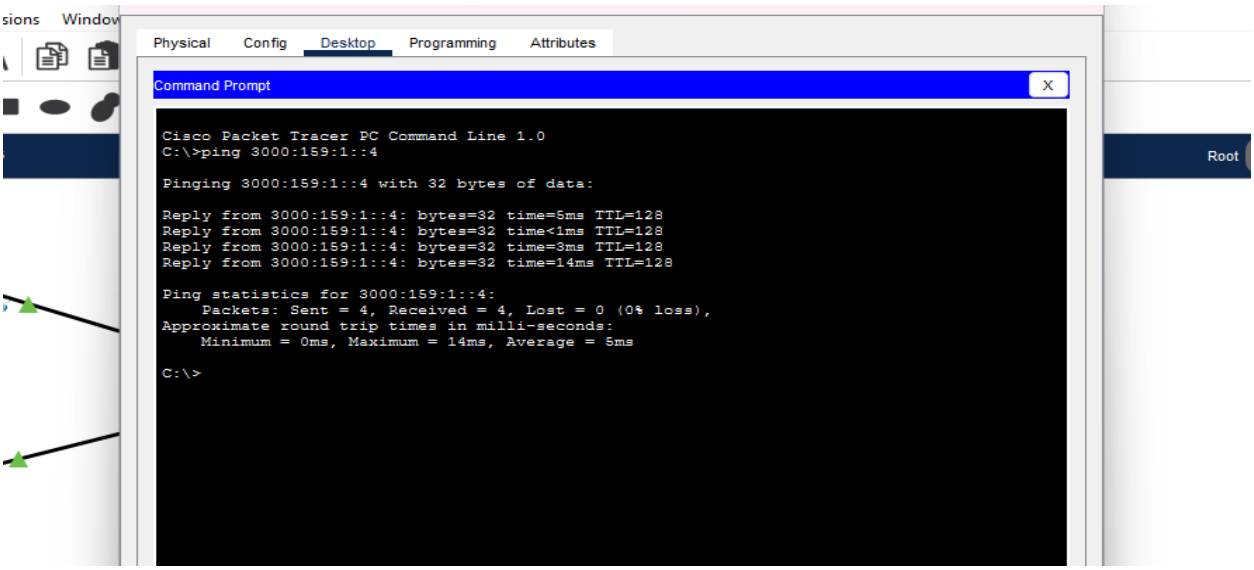

Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::2
/ 64
Link Local Address
FE80::20A:F3FF:FE08:6922
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
PC-PT Client1
PC-PT client2
Destination

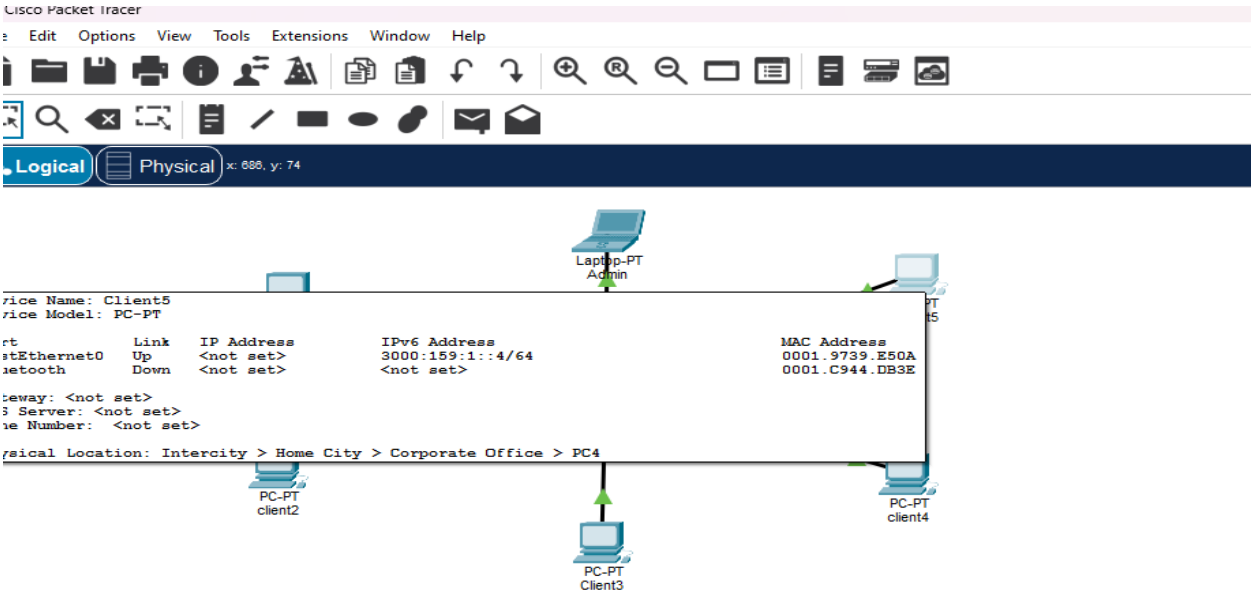

client2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time=3ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=26ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=29ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 29ms, Average = 14ms
C:\>
PC-PT Client1
PC-PT client2
Root

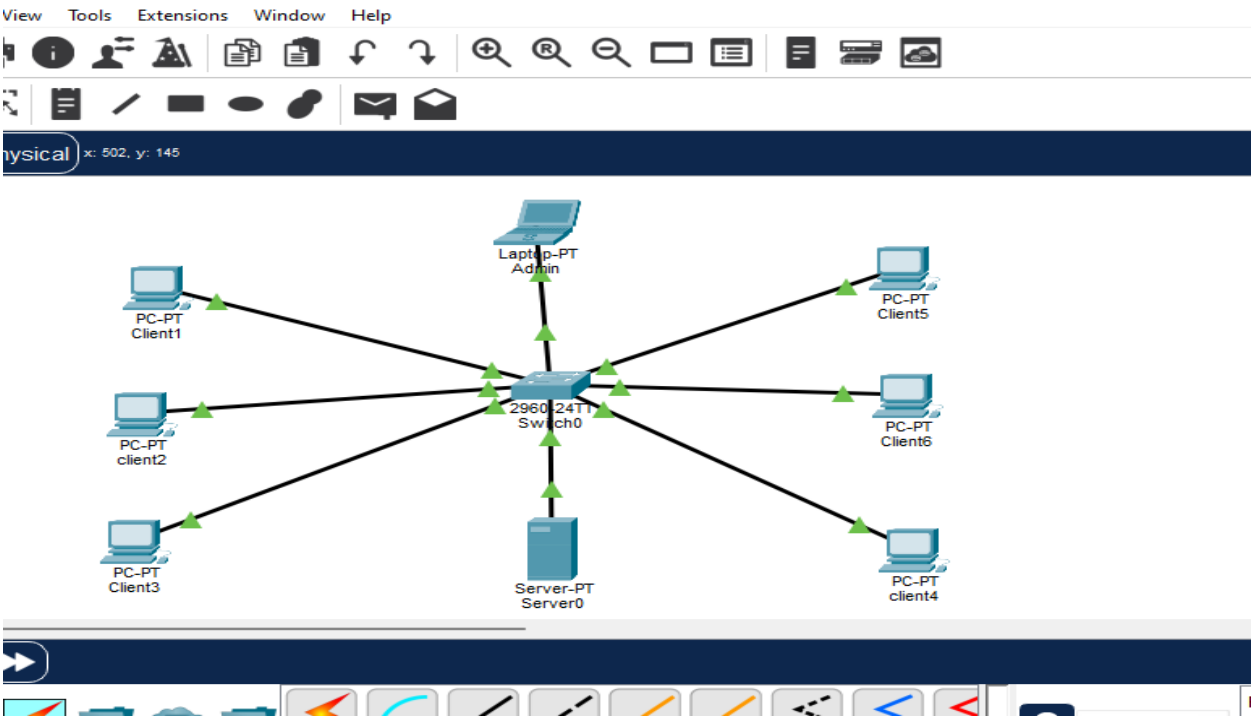

Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::201:97FF:FEEE:3330
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Root

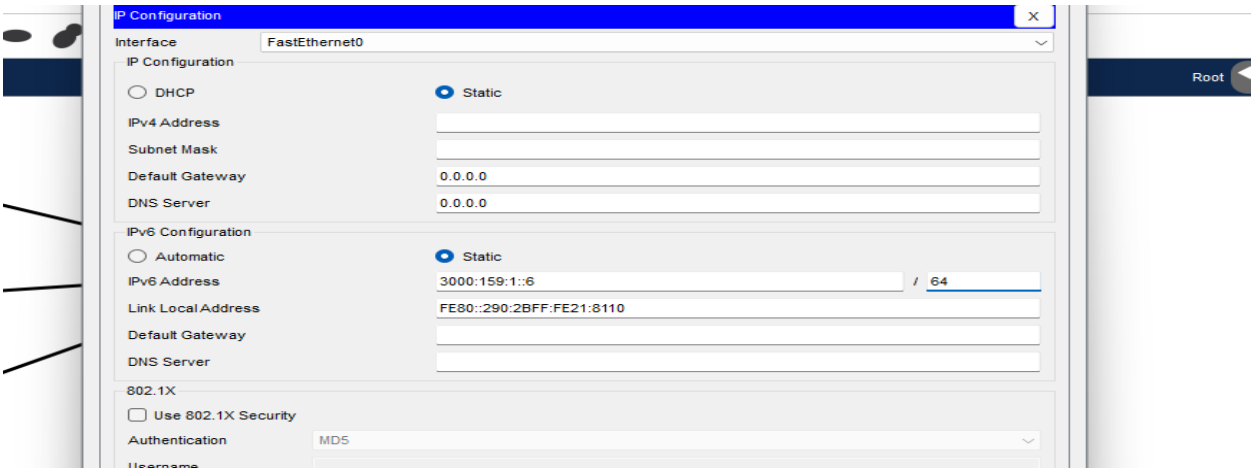

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::4
Pinging 3000:159:1::4 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 3000:159:1::4:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time=5ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=27ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 27ms, Average = 8ms
C:\>
Root

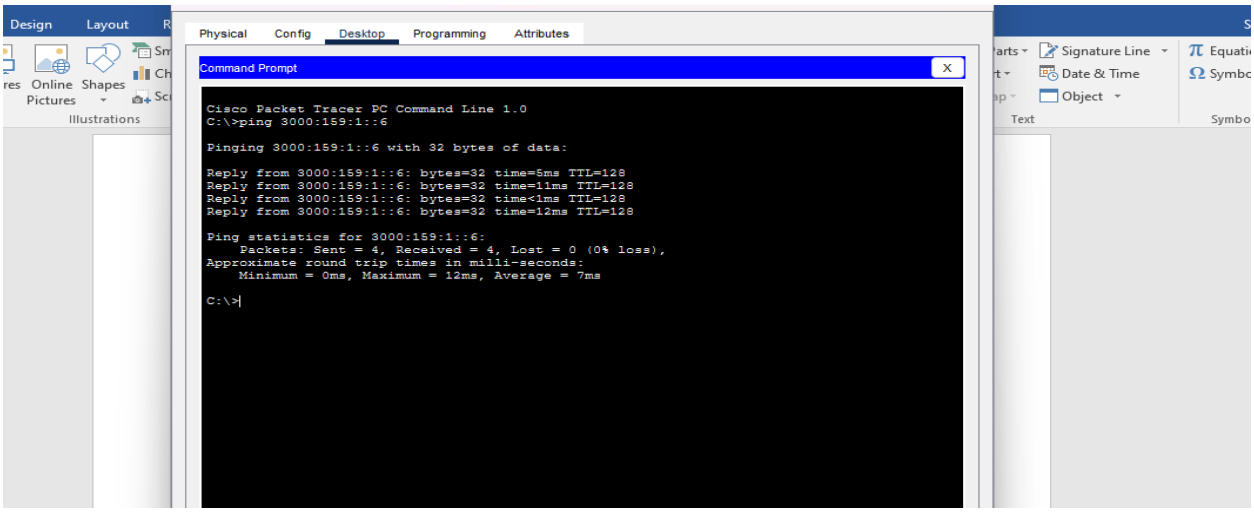

Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::4
/ 64
Link Local Address
FE80::210:11FF:FE07:4874
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Root
Destination
Type


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::4
Pinging 3000:159:1::4 with 32 bytes of data:
Reply from 3000:159:1::4: bytes=32 time=5ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=3ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=14ms TTL=128
Ping statistics for 3000:159:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 5ms
C:\>
Root

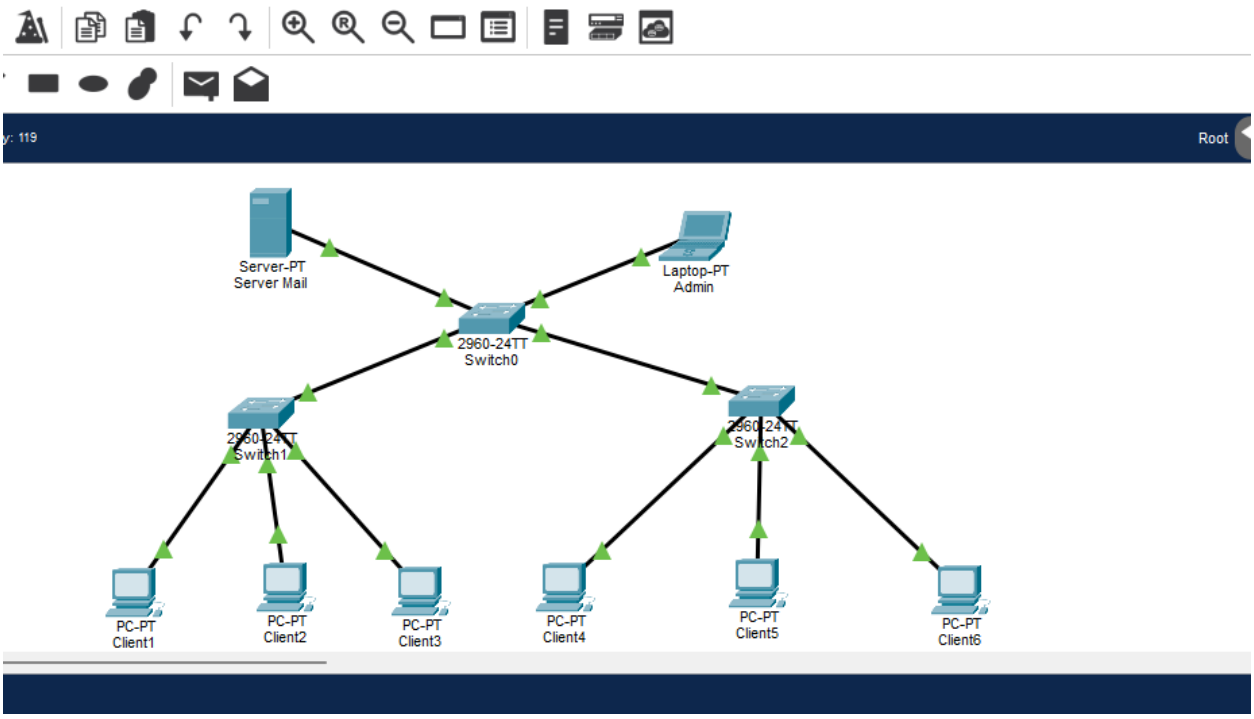

Cisco Packet Tracer
Edit Options View Tools Extensions Window Help
Logical
Physical
x: 688, y: 74
Laptop-PT
Admin
rice Name: Client5
rice Model: PC-PT
rt Link IP Address IPv6 Address MAC Address
stEthernet0 Up <not set> 3000:159:1::4/64 0001.9739.E50A
uetooth Down <not set> <not set> 0001.C944.DB3E
teway: <not set>
Server: <not set>
ne Number: <not set>
rsical Location: Intercity > Home City > Corporate Office > PC4
PC-PT
client2
PC-PT
Client3
PC-PT
client4


View Tools Extensions Window Help
ysical
x: 502, y: 145
Laptop-PT
Admin
PC-PT
Client1
PC-PT
Client5
PC-PT
client2
2960-24TT
Switch0
PC-PT
Client6
PC-PT
Client3
Server-PT
Server0
PC-PT
client4

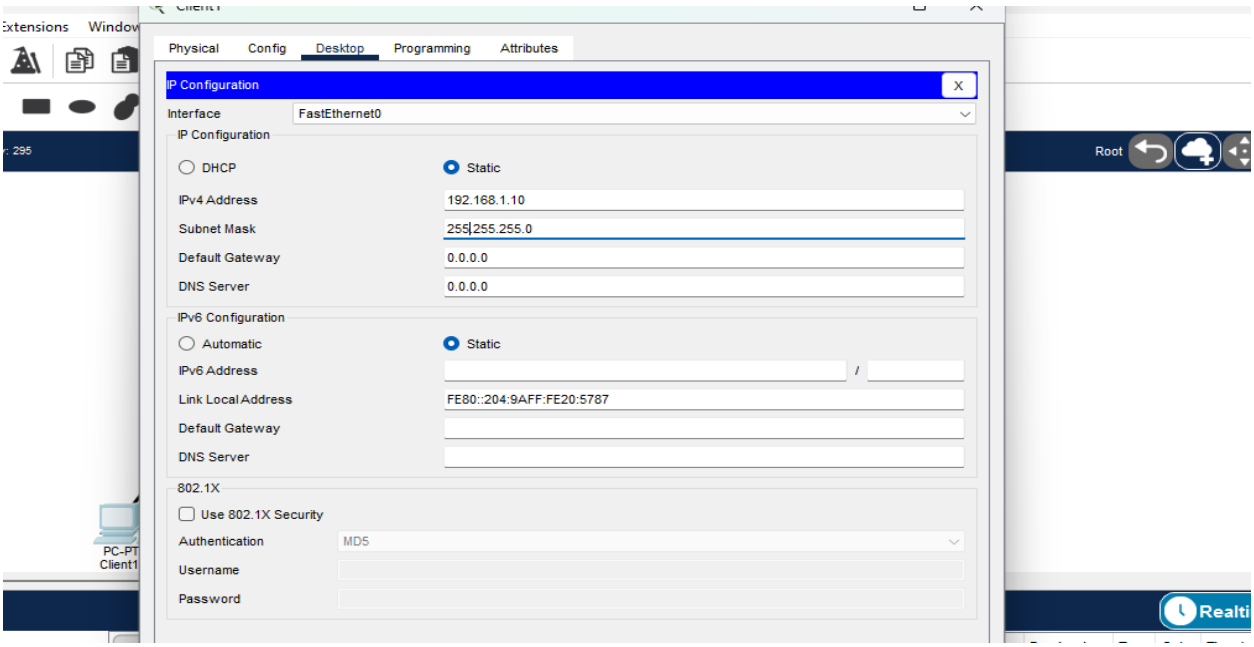

IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::6
/ 64
Link Local Address
FE80::290:2BFF:FE21:8110
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MDS
Root

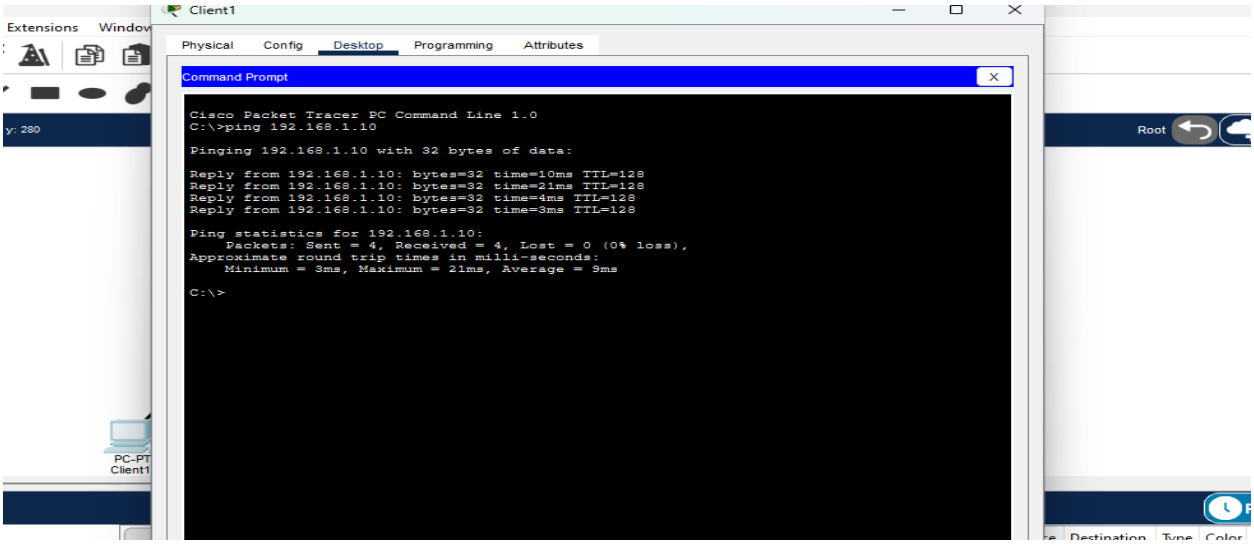

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::6
Pinging 3000:159:1::6 with 32 bytes of data:
Reply from 3000:159:1::6: bytes=32 time=5ms TTL=128
Reply from 3000:159:1::6: bytes=32 time=11ms TTL=128
Reply from 3000:159:1::6: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::6: bytes=32 time=12ms TTL=128
Ping statistics for 3000:159:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 7ms
C:\>
Design
Layout
Online Pictures
Shapes
Illustrations
Signature Line
Date & Time
Object
Text

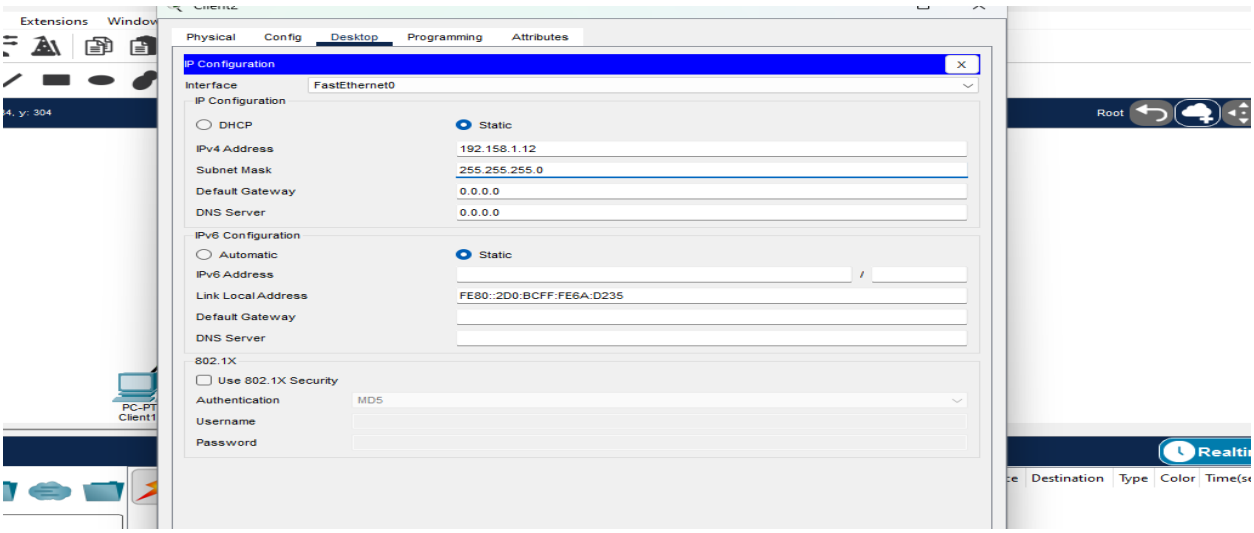
- Topologie en arbre.

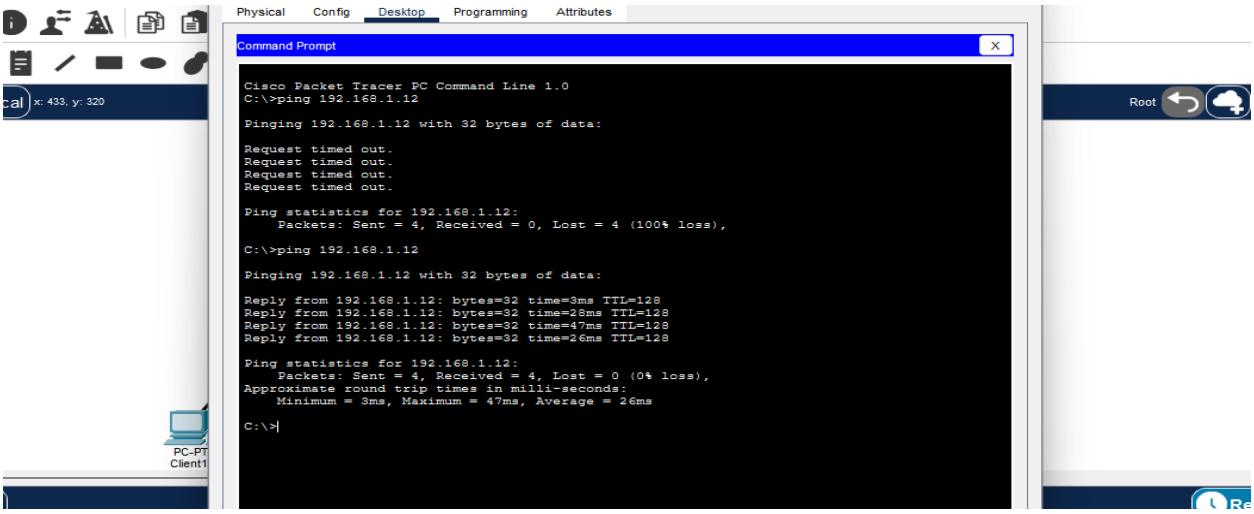


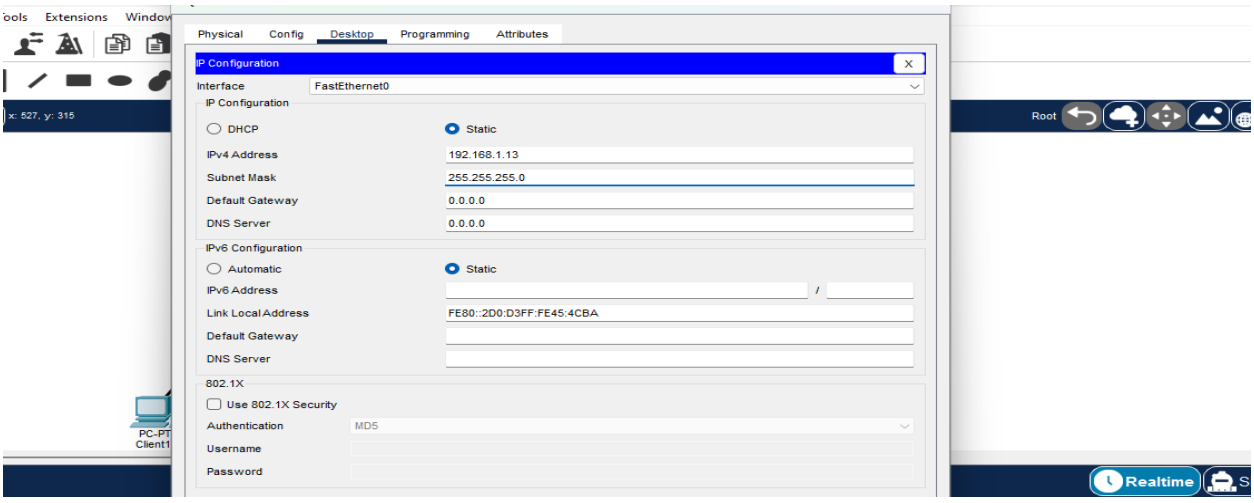
- Configuration des addresses IPV4 et testes de connectivité.

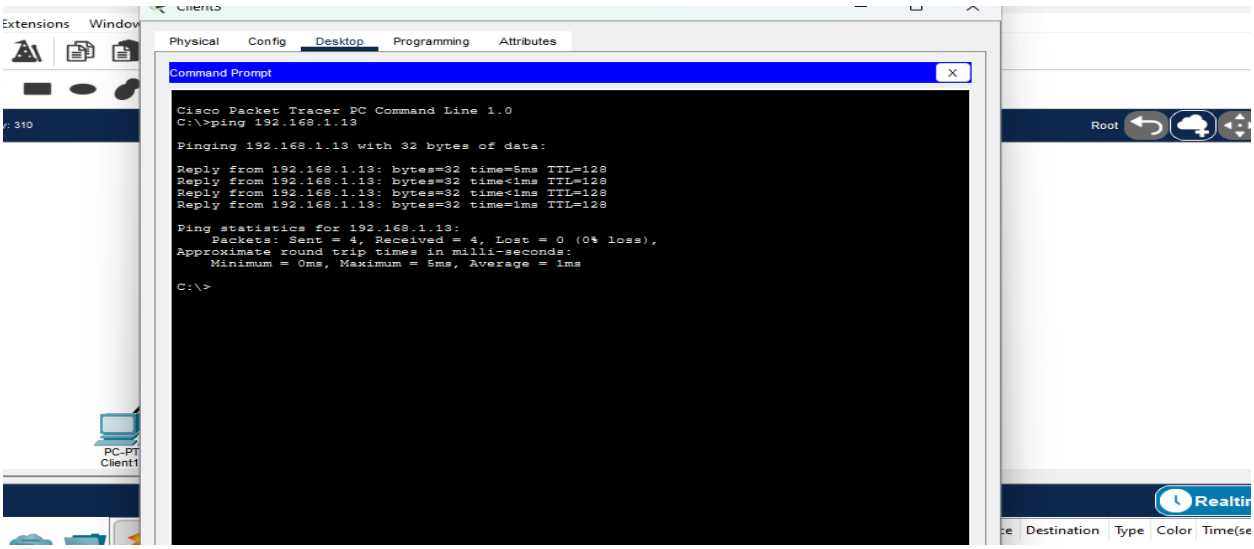


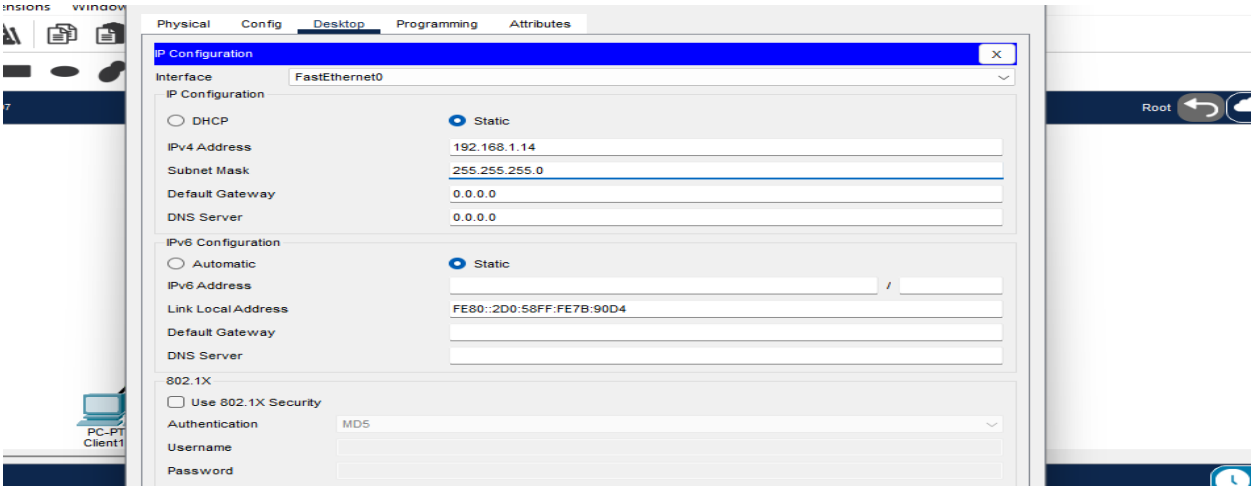

Client1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=10ms TTL=128
Reply from 192.168.1.10: bytes=32 time=21ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128
Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 21ms, Average = 9ms
C:\>
Root
PC-PT Client1

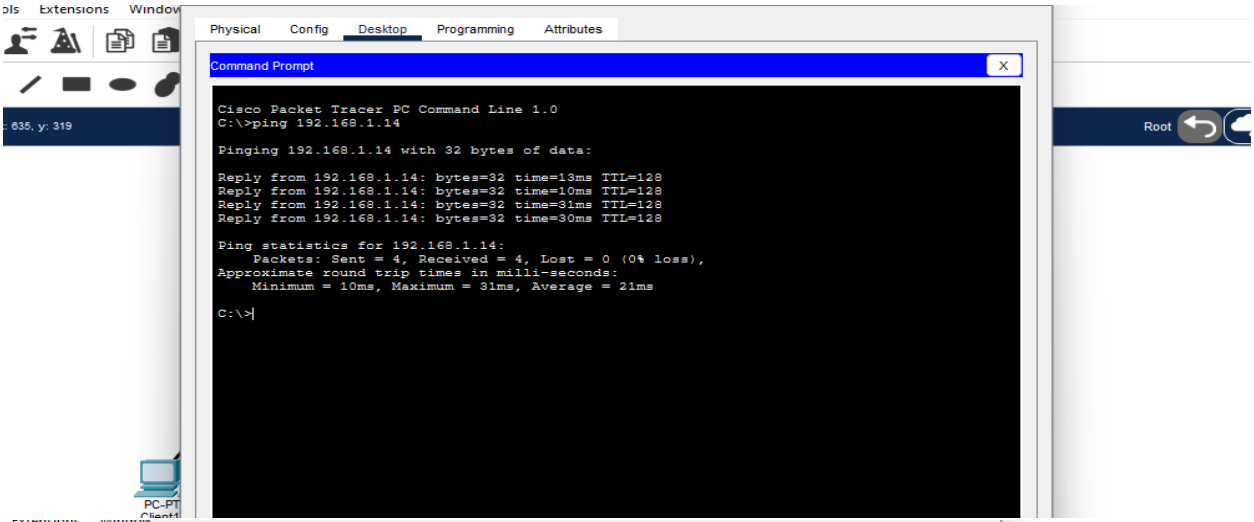

Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.158.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:BCFF:FE6A:D235
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Root
PC-PT Client1
Destination
Type
Color

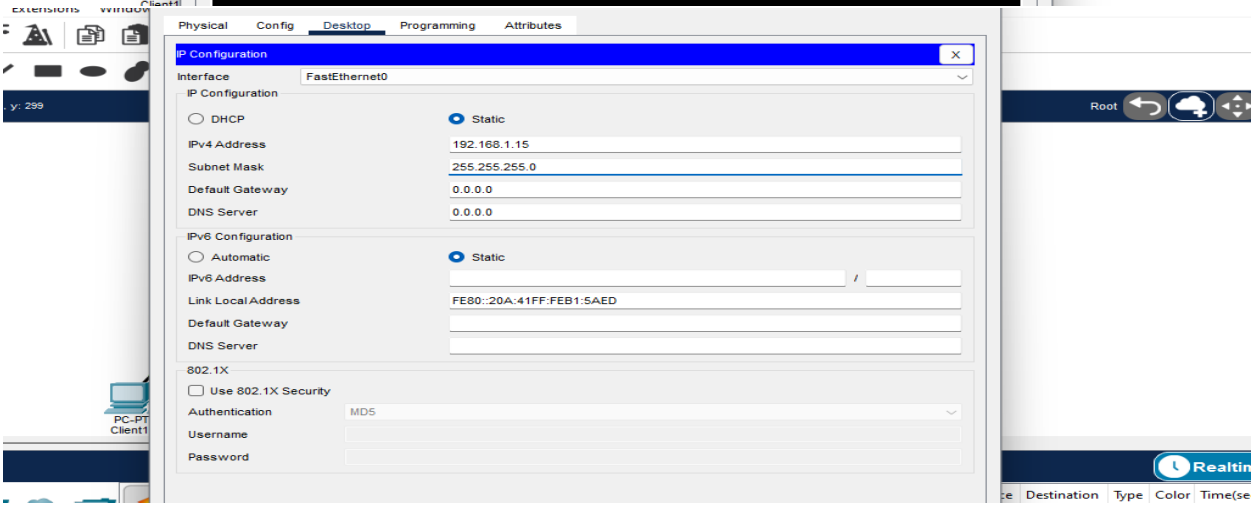

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time=3ms TTL=128
Reply from 192.168.1.12: bytes=32 time=28ms TTL=128
Reply from 192.168.1.12: bytes=32 time=47ms TTL=128
Reply from 192.168.1.12: bytes=32 time=26ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 47ms, Average = 26ms
C:\>
Root
PC-PT
Client1

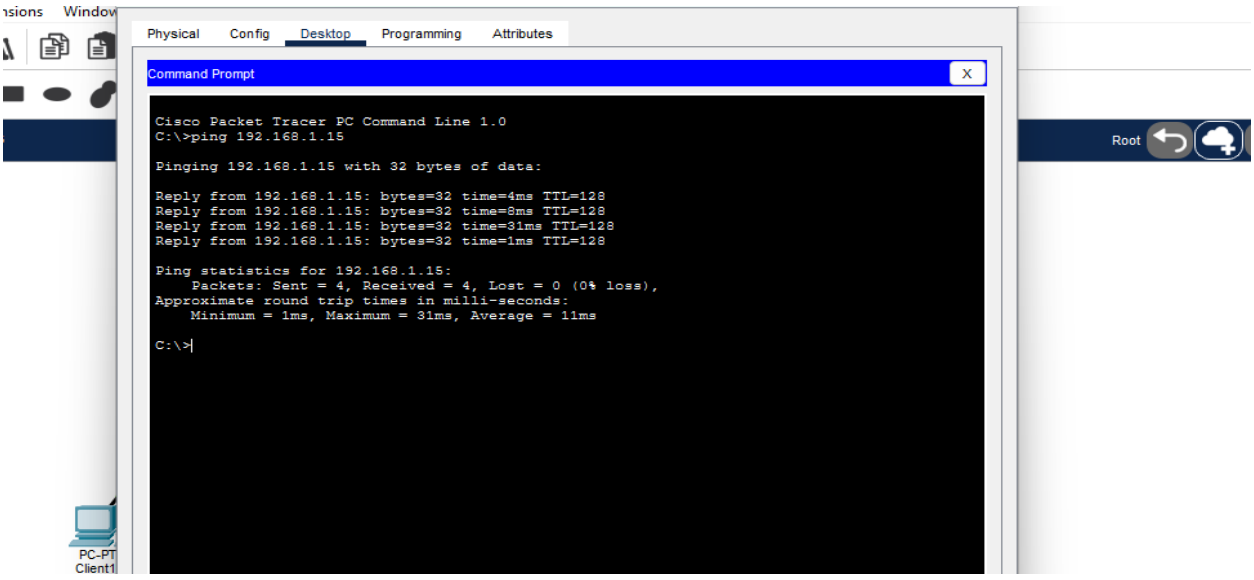

Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.13
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:D3FF:FE45:4CBA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Root
Realtime
PC-PT
Client1

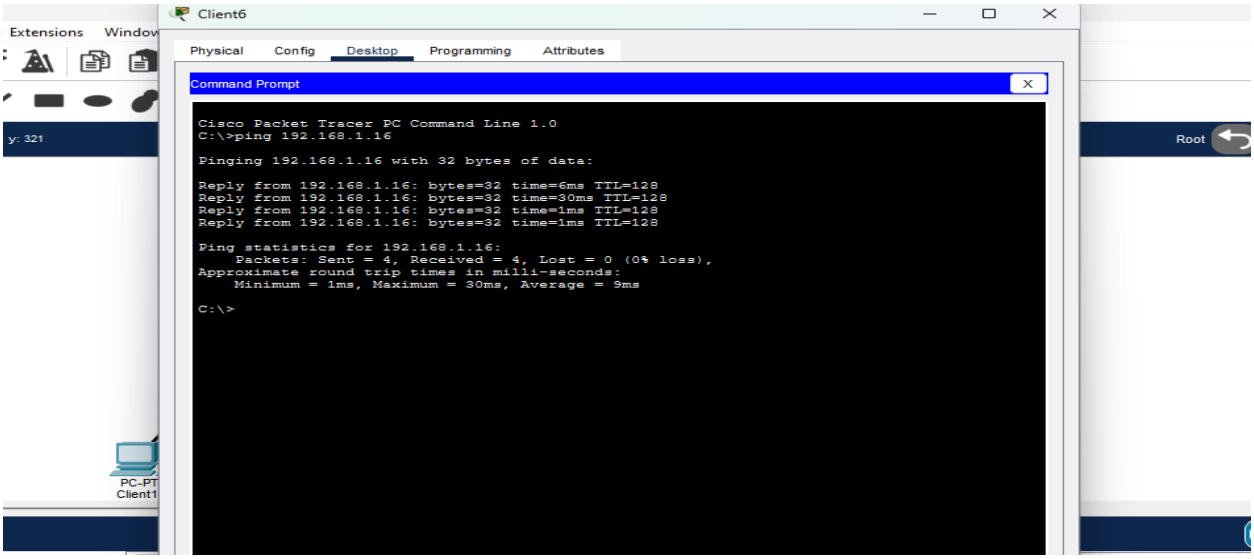

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time=5ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>
Root
PC-PT
Client1
Destination
Type
Color
Time(se


Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:58FF:FE7B:90D4
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Root
PC-PT
Client1

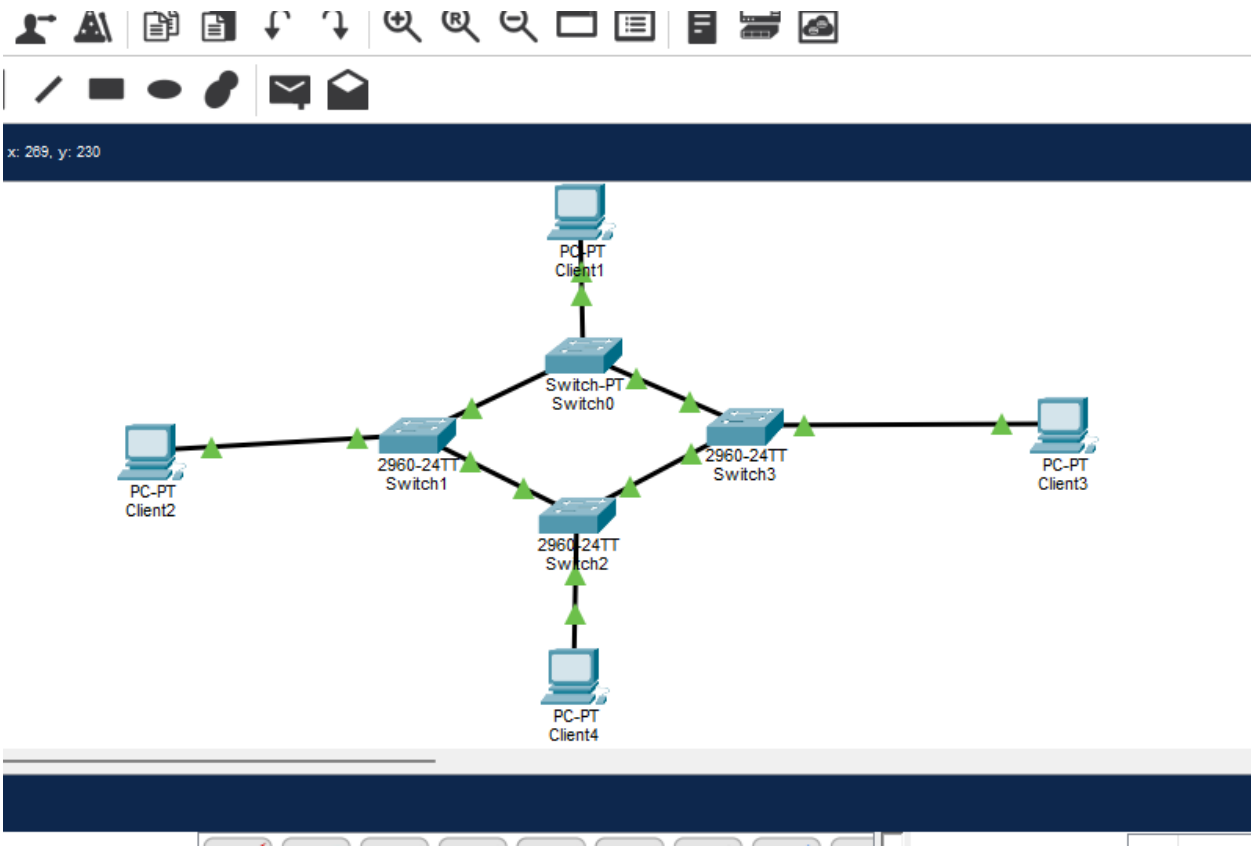

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.14
Pinging 192.168.1.14 with 32 bytes of data:
Reply from 192.168.1.14: bytes=32 time=13ms TTL=128
Reply from 192.168.1.14: bytes=32 time=10ms TTL=128
Reply from 192.168.1.14: bytes=32 time=31ms TTL=128
Reply from 192.168.1.14: bytes=32 time=30ms TTL=128
Ping statistics for 192.168.1.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 31ms, Average = 21ms
C:\>
Root
PC-PT
Client1


Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.15
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20A:41FF:FEB1:5AED
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Root
PC-PT
Client1
Destination
Type
Color
Time(se

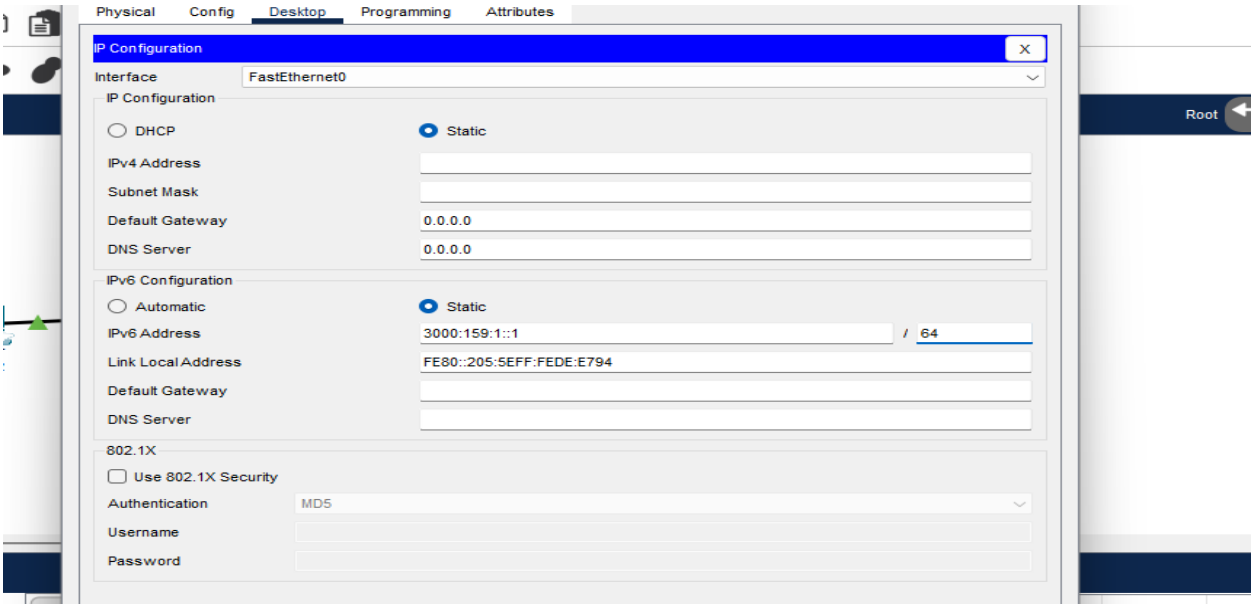

Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time=4ms TTL=128
Reply from 192.168.1.15: bytes=32 time=8ms TTL=128
Reply from 192.168.1.15: bytes=32 time=31ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 31ms, Average = 11ms
C:\>
Root
PC-PT
Client1

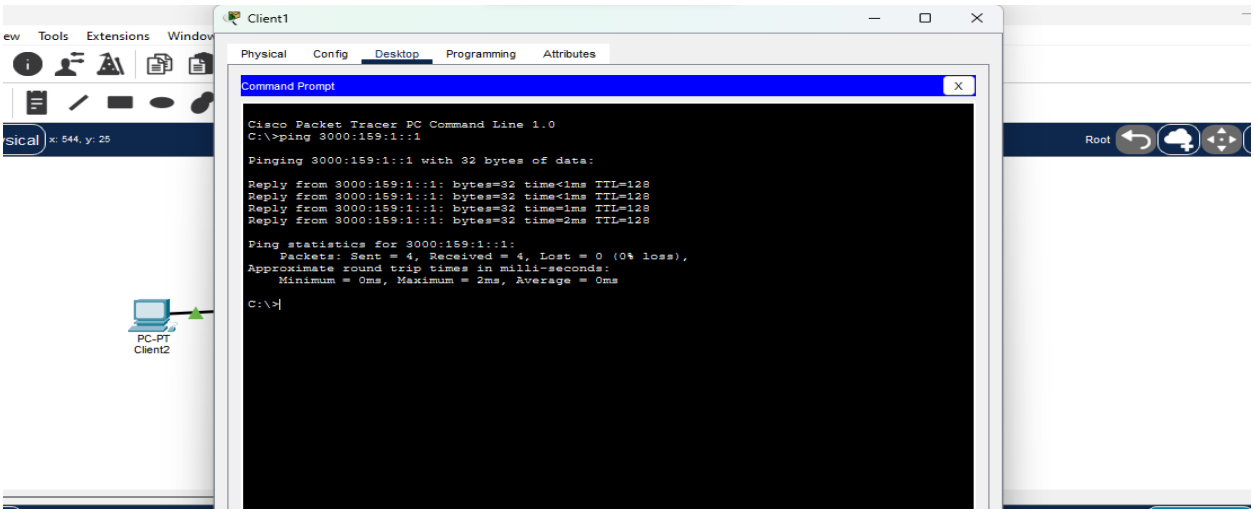

Client6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time=6ms TTL=128
Reply from 192.168.1.16: bytes=32 time=30ms TTL=128
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.16:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 30ms, Average = 9ms
C:\>
Root
PC-PT
Client1

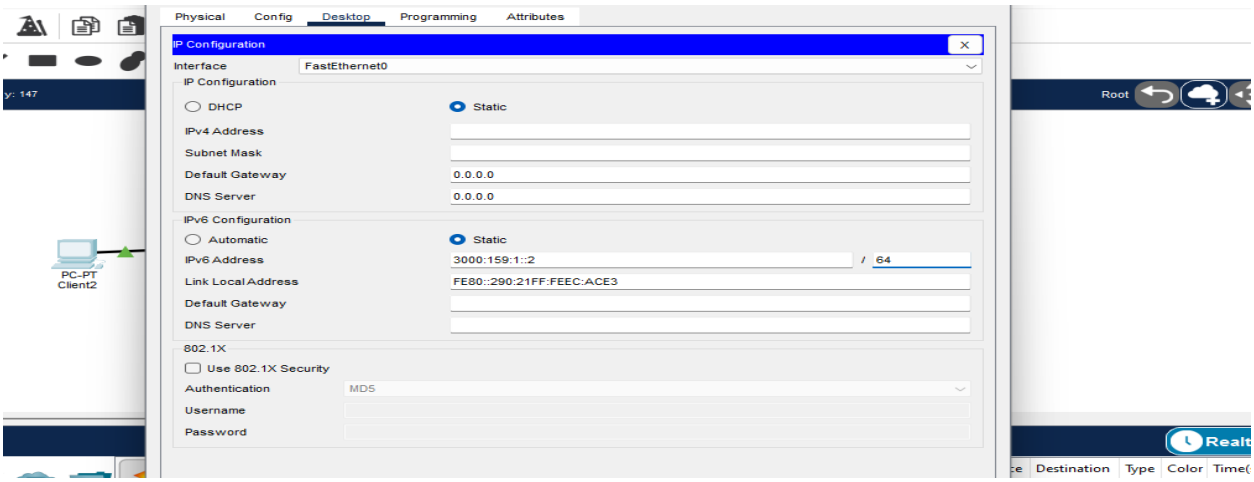
- Topologie en anneau.



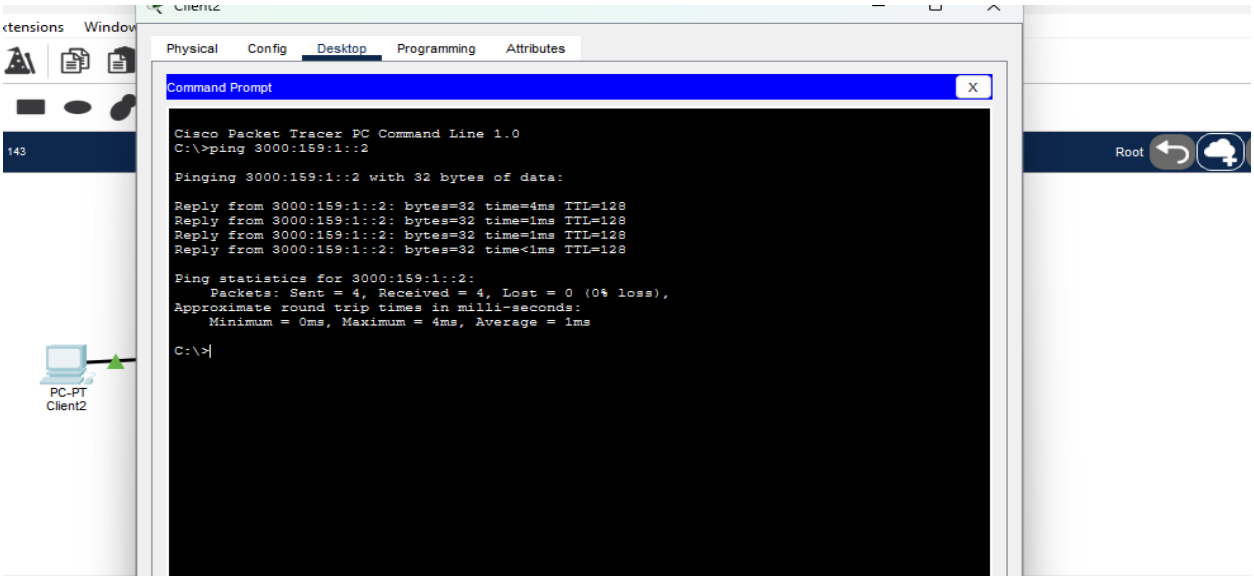
- Configuration des addresses IPV6 et testes de connectivité.

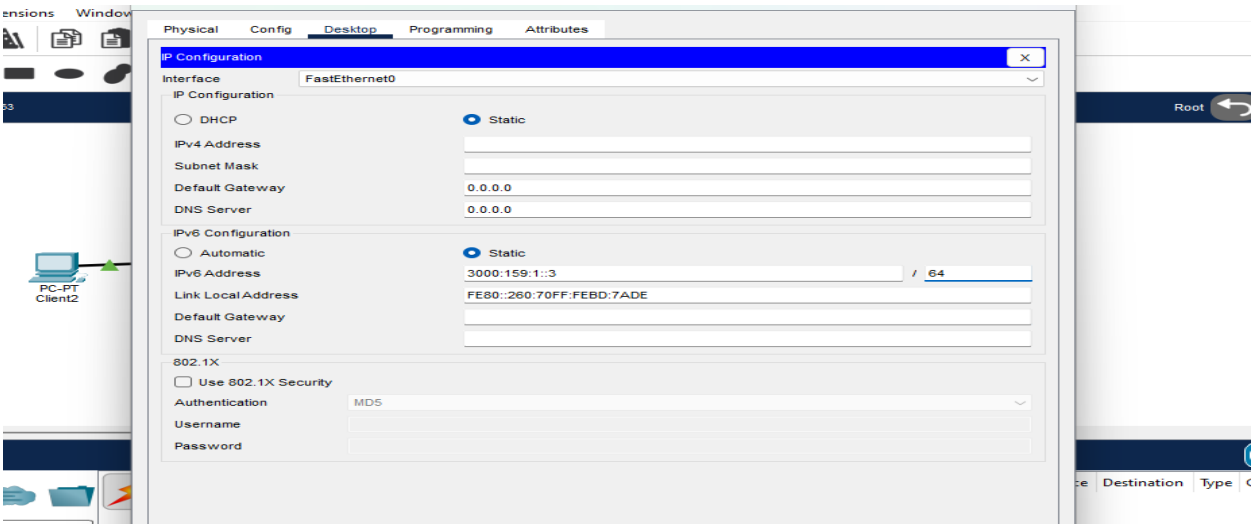


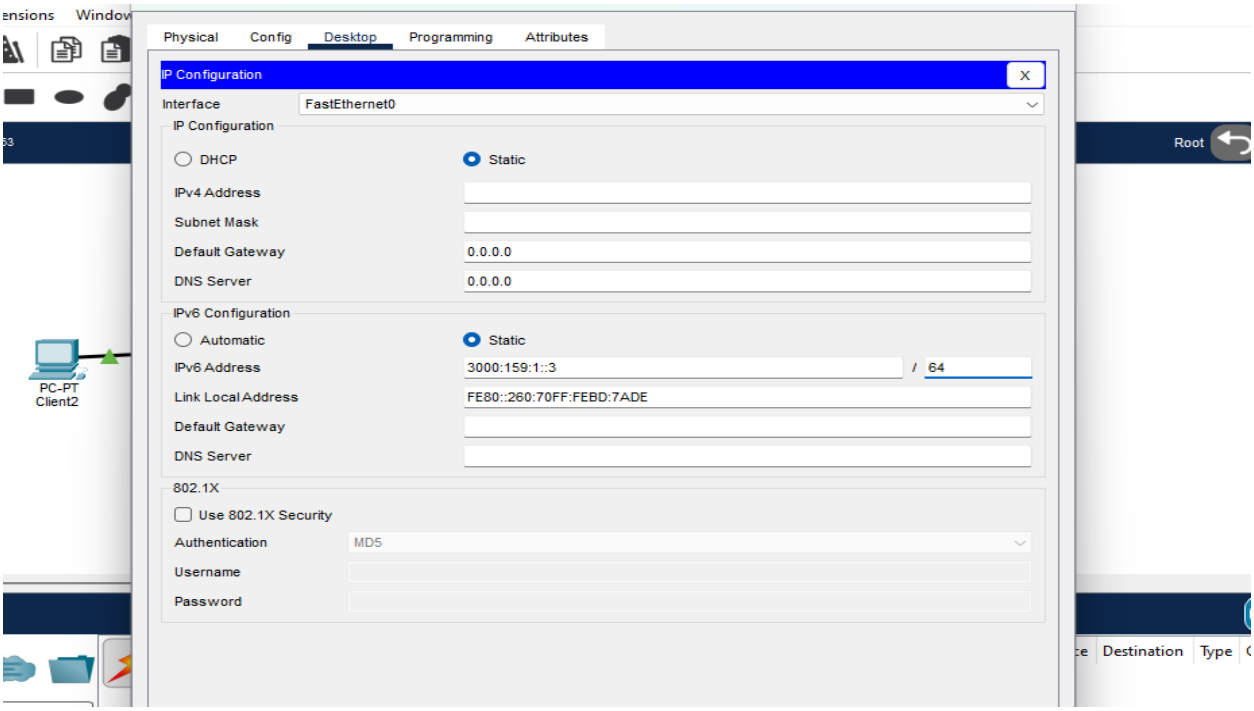




- Teste de connectivité avec ping et Adresse IPV6.




Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::260:70FF:FEBD:7ADE
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
PC-PT
Client2
Root
Destination
Type


Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::260:70FF:FEBD:7ADE
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
PC-PT
Client2
Root
Destination
Type

Démarche suivie

1 Création de la topologie de base : Mise en place d'équipements interconnectés (Hub ou Switch) et de plusieurs stations de travail (PC).

2 Configuration IPv4 : Accès à la configuration IP statique de chaque machine via l'onglet Desktop > IP Configuration de Cisco Packet Tracer pour y renseigner l'adresse IPv4 et le masque de sous-réseau correspondants.

3 Tests de connectivité IPv4 : Utilisation de l'invite de commandes (Command Prompt) pour exécuter la commande ping, afin de tester la communication d'un PC vers un autre.

4 Configuration IPv6 : Attribution d'adresses IPv6 statiques (ex: préfixe global 3000:159:1::X/64) sur les interfaces réseau des hôtes.

5 Tests de connectivité IPv6 : Vérification de la liaison à l'aide de la commande ping vers les adresses IPv6 configurées.

Conclusions

Résumé des apprentissages : Ce TD m'a permis de maîtriser la configuration manuelle des adresses IPv4 et IPv6 sur des équipements finaux via Cisco Packet Tracer, ainsi que l'utilisation des outils de diagnostic de base comme le ping pour valider les échanges de paquets.

Bilan, difficultés et solutions :

Difficultés rencontrées : Éventuelles erreurs de saisie dans les masques de sous-réseau ou oubli du format de préfixe /64 en IPv6.

Solutions apportées : Double vérification des tables d'adressage fournies dans l'énoncé et utilisation rigoureuse des fenêtres de configuration de Packet Tracer.

Statut : Tâche entièrement réussie avec une connectivité validée à 100% sur les topologies testées.